

Charlotte Sherratt

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Sydney, Australia

Interest in GIT: My recent research is on the GIT of a specific quiver variety, and more broadly I am interested in moduli of sheaves and vector bundles. I aim to pivot to applying this area of algebraic geometry to mathematical physics, given my undergraduate training in quantum field theory. In particular, I am interested in how objects I know from geometry (e.g. the quotient construction of the Hilbert scheme of points in $\mathbb{C}^2$) relate to physics to inform my future research and thesis directions.

Publication

Sole author. *The Geometric Invariant Theory of $4 \times 3 \times 3$ Tensors (Determinantal Cubic Surfaces)*.

In preparation, expected upload to the arXiv in April 2026. A draft can be found at the following link: <https://github.com/charlotte-sherratt/4x3x3-GIT-quotient/tree/main/paper-drafts>.

Education

Master 2 in mathematics

commencing 09/2026

Analysis, Number Theory, Geometry at the Université Paris-Saclay (Paris, France).

- Awarded a Sophie Germain scholarship. Awaiting confirmation of enrolment for the degree (this can be provided if requested).
- I am travelling in the year between degrees.

Bachelor of Philosophy (First Class Honours) – Science

02/2021–06/2025 (awarded)

Australian National University (Canberra, Australia).

An Australian degree at the level of Bachelor with Honours (equivalent to 240 ECTS) with a double major in mathematics and physics and 1 year of honours in mathematics.

- Average grade of 93/100 for the honours year, 88/100 for the bachelors degree. GPA 6.94/7.
- Recipient of a University Medal in 2025, awarded to the top two mathematics students for "academic excellence".

Honours thesis: *On determinantal representations of cubic surfaces*.

90 pages, written under the supervision of Dr Anand Deopurkar.

Work experience

Australian National University (Canberra, Australia)

Research assistant

09/2025–02/2026

- Continuing the research from my honours thesis.
- Involved the preparation of a publication on the GIT of determinantal representations of cubic surfaces (equivalently the GIT of a certain quiver variety) and the participation in a working group on Bridgeland stability.

Teaching assistant

02/2023–12/2025

- Instructor for workshops (tutorials) in mathematics (2024, 2025) and physics (2023, 2025). Responsible for grading homework and exams.
- Instructor for workshops for a course on PDEs taught at Shandong University, China.